

→ 4 Stroke Engine

4 stroke in each cylinder, 2 revolutions of crank shaft

→ Ideal gas

⊗ Egn of state: $PV = RT$
 $PV = mRT$

⊗ $\Delta h, \Delta u$: $U(T_2) - U(T_1) = \int_1^2 C_v(T) dT$

$h(T_2) - h(T_1) = \int_1^2 C_p(T) dT$

⊗ Constant specific heats

$\Delta u = U(T_2) - U(T_1) = C_v(T_2 - T_1)$

$\Delta h = h(T_2) - h(T_1) = C_p(T_2 - T_1)$

Table: Ideal gas specific heats

Variable specific heats

$U(T)$ & $h(T)$ from tables:
 Ideal gas properties of air.

⊗ ΔS / change in Entropy

$S(T_2, v_2) - S(T_1, v_1) = \int_1^2 C_v(T) \frac{dT}{T} + R \ln \frac{v_2}{v_1}$

Constant specific heats: $= C_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}$

$S(T_2, p_2) - S(T_1, p_1) = C_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$

⊗ $\Delta S = 0$: Equal entropy / Isentropic

$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{(k-1)/k}$

$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{k-1}$

$\frac{p_2}{p_1} = \left(\frac{v_1}{v_2} \right)^k$

$k = \frac{C_p}{C_v}$; from table

Variable specific heats

$S(T_2, p_2) - S(T_1, p_1) = S^\circ(T_2) - S^\circ(T_1) - R \ln \frac{p_2}{p_1}$

$S^\circ(T)$ from tables
 A-17

Variable sp. heats - air only

$\frac{p_2}{p_1} = \frac{Pr_2}{Pr_1}$

$\frac{v_2}{v_1} = \frac{vr_2}{vr_1}$

Air only
 Pr, vr from
 table of air
 A-17

Oil & Time $\rightarrow$ Production will be costly

$\rightarrow$ Fuel efficient car

- Light weight, high-strength materials,
- Automatic shuts, when car is idle

For isentropic process:

Air table

$$V_{r2} = V_{r1} \left(\frac{V_2}{V_1} \right) = \frac{V_{r1}}{r}$$

$$V_{r4} = V_{r3} \left(\frac{V_4}{V_3} \right) = r V_{r3}$$

Since: $V_3 = V_2$ & $V_4 = V_1$, $r = \frac{V_1}{V_2} = \frac{V_4}{V_3}$

V_r from table using Temperature

Otto cycle using cold air-standard basis:
 $T = 25^\circ\text{C}$, R_p , $C_v = C$

Isentropic: $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{k-1} = r^{k-1} \quad (k=C)$

$$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4} \right)^{k-1} = \frac{1}{r^{k-1}} \quad (k=C)$$

$$k = C_p / C_v$$

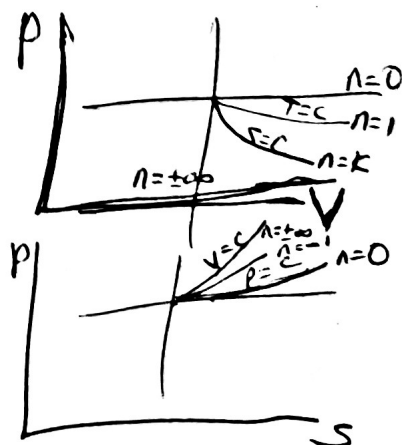
$PV^n = C \rightarrow$ Polytropic process

$PV^k = C \rightarrow$ Polytropic, Isentropic, ideal gas

$PV^n = C$: $n=1 \Rightarrow$ Isothermal ($T=C$)

$n=0 \Rightarrow$ Isobaric ($p=C$)

$n=\pm\infty \Rightarrow$ Isochoric ($V=C$)



Ideal gas model - Constant specific heats

$$U_2 - U_1 = C_v(T_2 - T_1)$$

$$h_2 - h_1 = C_p(T_2 - T_1)$$

$$S_2 - S_1 = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right) = C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{V_2}{V_1}\right)$$

$$S = f(T, P)$$

$$\text{or } f(T, V)$$

Isentropic process $s = \text{const}$

$$S_2 - S_1 = 0 = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)$$

$$\frac{C_p \ln\left(\frac{T_2}{T_1}\right)}{C_p} = \frac{R \ln\left(\frac{P_2}{P_1}\right)}{C_p}$$

$$\ln\left(\frac{T_2}{T_1}\right) = \frac{R}{C_p} \ln\left(\frac{P_2}{P_1}\right)$$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{R}{C_p}} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

$$\frac{T_2}{T_1} = \ln \left[\left(\frac{P_2}{P_1}\right)^{\frac{R}{C_p}} \right] = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

$$R = C_p - C_v$$

$$k = \frac{C_p}{C_v}$$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1}$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^k$$

$$(S_1 = S_2, \text{ideal gas, constant } k)$$

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Examples

1. Ideal Otto cycle

$r = 8$

~~At~~ ~~state~~

At state 1

$$P = 100 \text{ kPa}$$

$$T = 17^\circ\text{C}$$

$$\rho = 800 \text{ kg/m}^3$$

Processes:

- 1-2: adiabatic compression
- 2-3: $v=C$ heat addition
- 3-4: adiabatic expansion
- 4-1: $v=C$ heat rejection

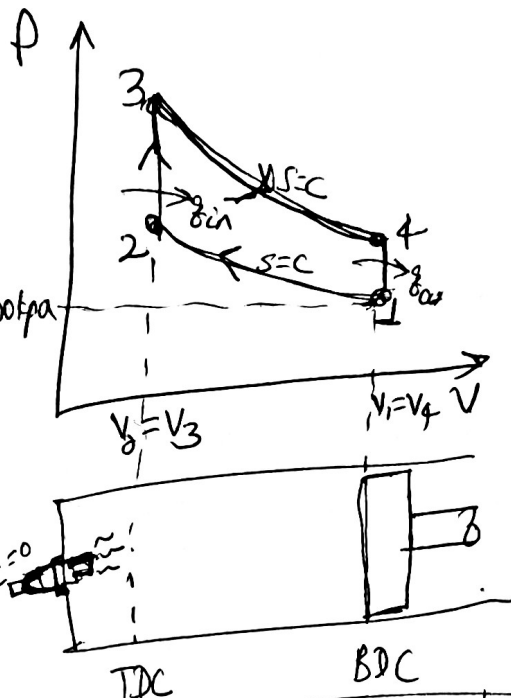
First law

$$U_2 - U_1 = Q_{12} - W_{12}$$

$$U_3 - U_2 = Q_{23} - W_{23}$$

$$U_4 - U_3 = Q_{34} - W_{34}$$

$$U_1 - U_4 = Q_{41} - W_{41}$$



Required

a. T_{max}

P_{max}

b. W_{net}

c. η_{th}

d. MEP

Assumptions

- The air-standard assumptions are applicable
- KE & PE are negligible.

$$\left(\frac{P_2}{P_1}\right) = \left(\frac{P_2}{P_1}\right) \cdot \frac{V_2}{V_1} = \frac{V_2}{V_1} \cdot \frac{V_1}{V_2}$$

Analysis

Isentropic process, ideal gas, variable specific heats with temp. considered

a. T_{max} & P_{max} are at point 3: T_3, P_3

at $T_1 = 17^\circ\text{C} = 290\text{K} \rightarrow V_1 = 676.1$

Process 1 $\rightarrow$ 2

Isentropic compression of ideal gas

$$\frac{V_2}{V_1} = \frac{V_2}{V_1} = \frac{1}{r}$$

$$\left(\frac{V_1}{V_2} = \frac{V_1}{V_2} = r\right)$$

$$P_1 V_1 = R T_1$$

$$P_2 V_2 = R T_2$$

Equation of state

$$\frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2}$$

$$\frac{P_1}{P_2} = \frac{T_1}{T_2} \cdot \frac{V_2}{V_1}$$

1st law

2 $\rightarrow$ 3

$$Q_{23} = U_3 - U_2$$

$$800 \text{ kJ/kg} = U_3 - 435.1124 \text{ kJ/kg}$$

$$U_3 = 1275.1124 \text{ kJ/kg}$$

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Everything with air-standard analysis and $T = 25^\circ\text{C}$, $c_p, c_v = \text{const}$

2. Cold-standard assumptions considered:

$$T = 25^\circ\text{C} : c_p, c_v = \text{const}$$

$$c_p =$$

$$c_v =$$

$$U_2 - U_1 = c_v (T_2 - T_1)$$

$$h_2 - h_1 = c_p (T_2 - T_1)$$

	$P(\text{kPa})$	$T(\text{K})$	U
1			
2			
3			
4			

$$\eta = \frac{W_{\text{net}}}{Q_{23}}$$

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$$P_2 V_2 = R T_2 \rightarrow R = \frac{P_2 V_2}{T_2} = \frac{P_1 V_1}{T_1}$$

$$P_1 V_1 = R T_1$$

$$P_2 V_2 T_2 = P_1 V_1 T_1$$

$$P_2 = \left(\frac{V_1}{V_2} \right) \left(\frac{T_1}{T_2} \right) P_1$$

$$= r \left(\frac{T_1}{T_2} \right) P_1$$

$$= 8 * \left(\frac{652.4}{290K} \right) * 100kPa$$

$$\text{Process 2-3} = \frac{1799.7 kPa}{U}$$

$$\frac{V_r}{6.301} \quad U = 1260.99$$

$$? \quad U = 1275.11$$

$$6.046 \quad U = 1279.65$$

$$\frac{? - 1560}{1580 - 1560} = \frac{1275.11 - 1260.99}{(1279.65 - 1260.99)}$$

$$? = 1575.134$$

$$T_3 = 1575.134$$

$$V_{r3} = 6.108$$

$$\text{Can of state: } \frac{P_3 V_3}{P_2 V_2} = \frac{R T_3}{R T_2}; V_3 = V_2$$

$$\rightarrow \frac{P_3}{P_2} = \frac{T_3}{T_2}$$

$$P_3 = \left(\frac{T_3}{T_2} \right) P_2 = 4.375 \text{ MPa}$$

$$\frac{P_3}{R T_3} = \frac{P_2}{R T_2}$$

$$\text{or } \frac{P_3}{T_3} = \frac{P_2}{T_2}$$

$$P_3 T_2 = P_2 T_3$$

$$\frac{P_3}{P_2} = \frac{T_3}{T_2}$$

For cycle:

$$W_{net} = Q_{net} = Q_{in} - Q_{out}$$

$$Q_{in} = 800 \text{ kJ/p}$$

$$Q_{out} = ?$$

$$- Q_{out} = \dots$$

$$- Q_{out} = U_1 - U_4$$

$$Q_{out} = U_4 - U_1$$

$$U_4 = ?$$

Process 3-4: Isentropic expansion of ideal gas

$$r = \frac{V_1}{V_2} = \frac{V_4}{V_3}$$

$$\frac{V_{r4}}{V_{r3}} = \frac{V_4}{V_3} = r \Rightarrow V_{r4} = r V_{r3} = 8 * 6.108 = 48.864$$

$$\text{Using } V_{r4} \Rightarrow T_4 = 795.70$$

$$\begin{array}{ccc} U & & \\ \frac{V_r}{V_r} & & \\ 51.64 & & 576.42 \\ 48.864 & & ? \\ 48.08 & & 592.30 \end{array}$$

$$800$$

$$\frac{? - 780}{20} = \frac{48.864 - 51.64}{48.08 - 51.64}$$

$$? = 795.70 K$$

$$U = 588.907$$

$$\begin{aligned} \Rightarrow q_{out} &= U_4 - U_1 \\ &= 588.74 - 206.91 \\ &= \underline{\underline{381.83 \text{ kJ/kg}}} \end{aligned}$$

$$\begin{aligned} \text{b. } \Rightarrow W_{net} &= q_{net} = q_{in} - q_{out} \\ &= (800 - 381.83) \text{ kJ/kg} \\ &= \underline{\underline{418.17 \text{ kJ/kg}}} \end{aligned}$$

$$\begin{aligned} \text{c. } \eta_{th} &= \frac{W_{net}}{q_{in}} \\ &= \frac{418.17 \text{ kJ/kg}}{800 \text{ kJ/kg}} = 0.523 \text{ or } \underline{\underline{52.3\%}} \end{aligned}$$

Checking with $\eta_{m, Otto} = 1 - \frac{1}{r^{k-1}}$
 $= 1 - \frac{1}{8^{1.4-1}}$ can't use since cp & cv are variable
 $\neq 56.5\%$

$$\begin{aligned} P_2 &= \frac{P_1}{r^k} = \frac{100 \text{ kPa}}{8^{1.4}} = 6.76 \text{ kPa} \\ T_2 &= T_1 \cdot \frac{V_1^{k-1}}{V_2^{k-1}} = 290 \text{ K} \cdot \frac{1}{8^{1.4-1}} = 84.51 \text{ K} \\ \left(\frac{V_2}{V_1}\right) &= \frac{V_2}{V_1} = \frac{1}{8} \\ \left(\frac{1}{r}\right) V_1 &= V_2 \\ V_2 &= 0.103996 \text{ m}^3/\text{kg} \\ \text{MEP} &= \frac{418.17}{0.832 - 0.103996} \\ &= \underline{\underline{574.409 \text{ kPa}}} \end{aligned}$$

d. The mean effective pressure:

$$\text{MEP} = \frac{W_{net}}{V_1 - V_2} = \frac{W_{net}}{V_1 - \frac{V_1}{r}} = \frac{W_{net}}{V_1 \left(1 - \frac{1}{r}\right)}$$

$$\text{where } V_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}) \cdot (290 \text{ K})}{100 \text{ kPa}}$$

$$= \underline{\underline{0.832 \text{ m}^3/\text{kg}}}$$

$$\begin{aligned} \text{MEP} &= \frac{418.17 \text{ kJ/kg}}{(0.832 \text{ m}^3/\text{kg}) \left(1 - \frac{1}{8}\right)} \cdot \left(\frac{1 \text{ kPa} \cdot \text{m}^3}{1 \text{ kJ}}\right) \\ &= \underline{\underline{574 \text{ kPa}}} \end{aligned}$$

1. A constant pressure of 574 kPa during power stroke would produce the same net work output as the entire cycle.

Cold Air standard Assumptions: 2-continued

At 300K = 27°C

Table A-2

$C_p = 1.005 \text{ kJ/kg} \cdot \text{K}$

$C_v = 0.718 \text{ kJ/kg} \cdot \text{K}$

$T_1 = 17^\circ\text{C}$
 $P_1 = 100 \text{ kPa}$

$q_{in} = 800 \text{ kJ/kg}$
 $r = 8$

Process 1-2: I s e n t r o p i c c o m p r e s s i o n :

↳ Isentropic relations (Adiabatic, Reversible, Internally)

$W_{12} = U_2 - U_1$

$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1} = (r)^{k-1}$

where $r = \frac{V_1}{V_2} = \frac{V_4}{V_3}$

⊗ $T_2 = \underline{\hspace{2cm}}$

$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^k = (r)^k$

or Eqn of state

$P_1 V_1 = RT_1 \rightarrow R = \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$P_2 V_2 = RT_2$

⊗ $P_2 = \underline{\hspace{2cm}}$

$P_2 = \frac{P_1 V_1 T_2}{T_1 V_2} = \frac{P_1}{T_1} \left(\frac{V_1}{V_2}\right) T_2$

$P_2 = P_1 \left(\frac{T_2}{T_1}\right) r$

⊗ $\frac{P_3 V_3}{T_3} = \frac{P_2 V_2}{T_2}$

$q_{in} = q_{2-3} = U_3 - U_2$

$q_{2-3} = C_v (T_3 - T_2)$

$800 \text{ kJ/kg} = 0.718 \text{ kJ/kg} \cdot \text{K} (T_3 - T_2)$

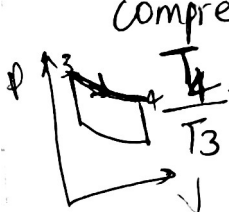
⊗ $P_3 = P_2 \left(\frac{T_3}{T_2}\right)^{\frac{1}{k}}$

⊗ $T_3 = \underline{\hspace{2cm}}$

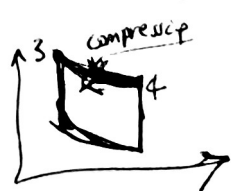
Process 3-4

Expansion

so go up to 3-4 since it is not compressing.
compression ratio will be $\frac{1}{r}$



$\frac{T_4}{T_3} = \left(\frac{1}{r}\right)^{k-1}$ or $\frac{T_3}{T_4} = (r)^{k-1}$



$\frac{P_4 V_4}{T_4} = \frac{P_3 V_3}{T_3} \Rightarrow P_4 = P_3 \left(\frac{T_4}{T_3}\right)^{\frac{1}{k}} \left(\frac{V_3}{V_4}\right)^{\frac{1}{k}}$

$\Delta U = q - w$

1-2: $U_2 - U_1 = -W_{12} = -C_v(T_2 - T_1)$

2-3: $U_3 - U_2 = q_{23}$

$C_v(T_3 - T_2) = q_{23}$

3-4: $U_4 - U_3 = -W_{34} = -C_v(T_4 - T_3)$

⊗ $W_{net} = W_{12} + W_{23} + W_{34} + W_{41}$

$(-ve) \quad (ve) \quad (ve) \quad (ve)$

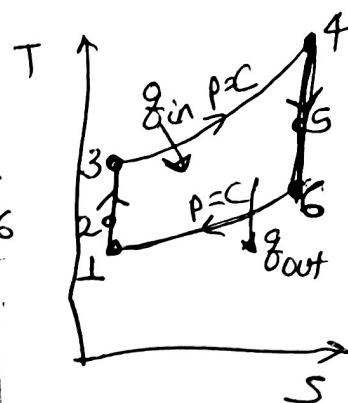
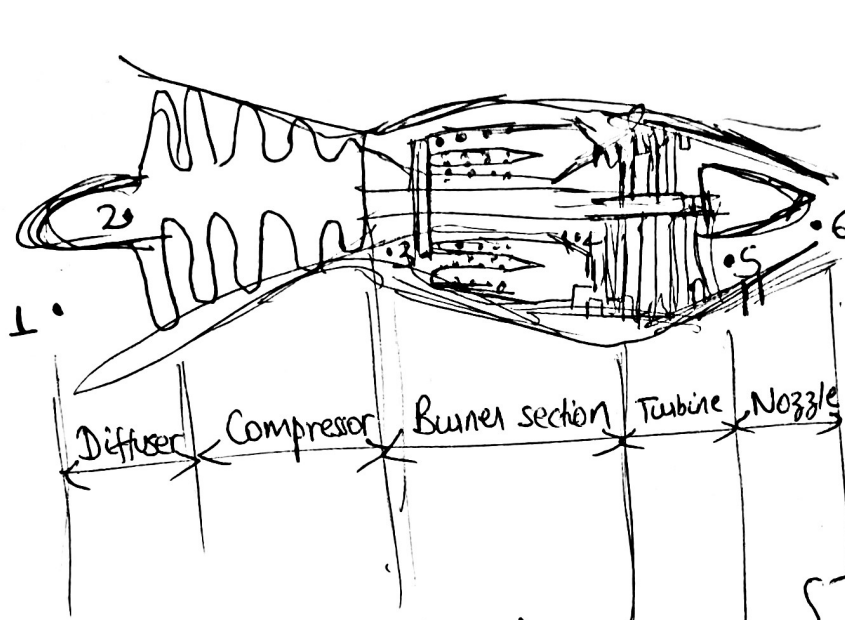
$= -C_v(T_2 - T_1) + W_{34}$

⊗ $\eta_{th} = \frac{W_{net}}{q_{23}} = \frac{1}{1 - \frac{T_2}{T_3}} = \frac{1}{1 - \frac{1}{r^{k-1}}} = \frac{r^{k-1} - 1}{r^{k-1}}$

$\eta_{th, Otto} = 1 - \frac{1}{r^{k-1}} = 1 - \frac{1}{8^{1.4}} = 56.5\% = 0.565$

Examples

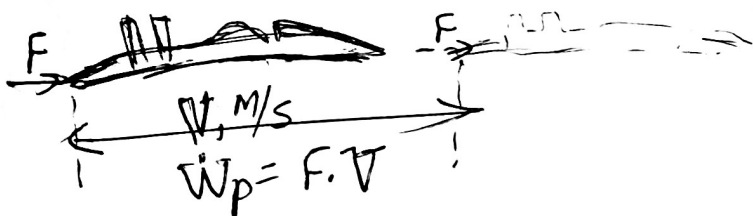
Turbojet Engines



Where = 0

$$F = (\dot{m} V)_{\text{exit}} - (\dot{m} V)_{\text{inlet}} \\ = \dot{m} (V_{\text{exit}} - V_{\text{inlet}}) \quad (N)$$

Relative to Aircraft $\left\{ \begin{array}{l} V_{\text{inlet}} - \text{Inlet velocity of air} \\ V_{\text{exit}} - \text{Exit velocity of exhaust gases} \end{array} \right.$
 $\dot{m}$ - mass flow rate of air through the engine



Propulsive power, $\dot{W}_p$
 $\downarrow$ Propulsive force / Thrust

$$\dot{W}_p = F * \frac{\text{distance this force acts on the aircraft}}{\text{Time}} \\ = \frac{\text{Thrust} * V_{\text{aircraft}}}{\text{Time}} \\ = \dot{m} (V_{\text{exit}} - V_{\text{inlet}}) V_{\text{aircraft}} \quad (kW)$$

$\dot{W}_p$ - power produced to propel the aircraft
 $\uparrow$ Desired output

Q_{in} - Heating value of the fuel

$$\eta_p = \frac{\text{Propulsive power}}{\text{Energy input rate } Q_{\text{in}}}$$

$\uparrow$ How efficiently the thermal energy released during combustion process is converted to propulsive energy.

Polytropic process

$$(q_{in} - q_{out}) + (w_{out} - w_{in}) = \Delta U$$

$\rightarrow = 0$ Isentropic

$\rightarrow \neq 0$ Polytropic

$$PV^n = C$$

$$W = \int_{in}^{final} P dV = \int_{in}^{final} \frac{C}{V^n} dV \quad \left| P = \frac{C}{V^n} \right.$$

$$= C \int_{in}^f \frac{1}{V^n} dV$$

$$= \frac{1}{n-1} (P_f V_f - P_i V_i)$$

$$= \frac{1}{n-1} \left(\frac{RT_f}{V_f} - \frac{RT_i}{V_i} \right)$$

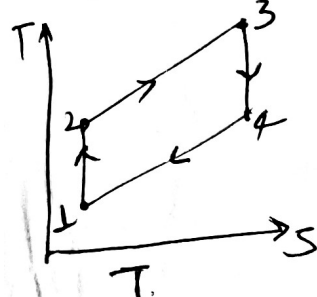
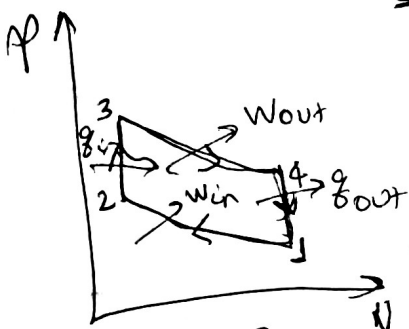
$$W = \frac{R}{n-1} (T_f - T_i)$$

$$\Rightarrow W = C \int_i^f V^{-n} dV = C \left[\frac{V_f^{-n+1} - V_i^{-n+1}}{-n+1} \right]$$

but $PV^n = C$

$$= \frac{P_f V_f^{-n+1} - P_i V_i^{-n+1}}{-n+1}$$

$$= \frac{P_f V_f - P_i V_i}{1-n}$$



$$PV = RT$$

$$P = \frac{RT}{V}$$

$$W_{in} = W_{12} = \frac{R}{n-1} (T_2 - T_1)$$

$$W_{out} = W_{34} = \frac{R}{n-1} (T_4 - T_3)$$

$$W_{net} = W_{out} - W_{in}$$

Isentropic: $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$

Polytropic $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{n-1}$

$$T_2 = T_1 (r)^{n-1} ; T_4 = T_3 \left(\frac{1}{r} \right)^{n-1}$$

* $\eta = \frac{W_{net}}{q_{in}} \rightarrow q_{2-3}$

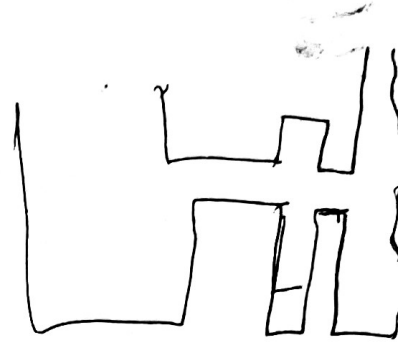
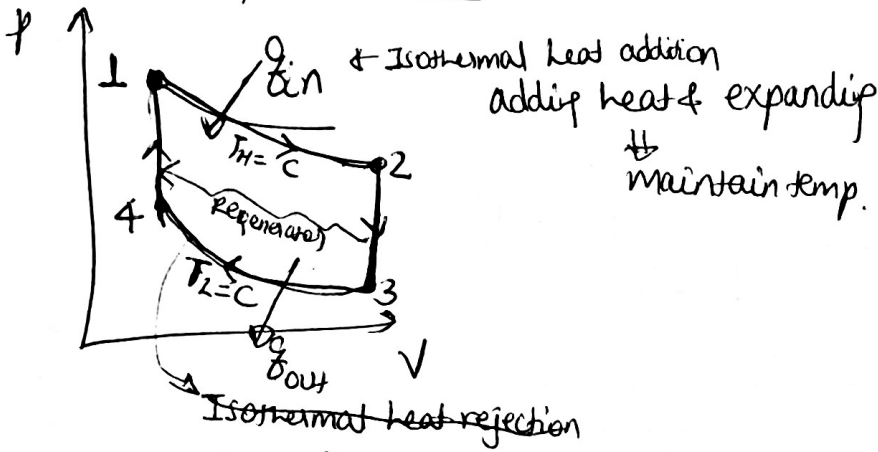
* $MEP = \frac{W_{net}}{V_1 - V_2}$

State 1
 $P_1 =$
 $T_1 =$
 $V_1 = \frac{RT_1}{P_1} =$

State 2
 $T_2 = T_1 (r)^{n-1}$
 $T_4 = T_3 \left(\frac{1}{r} \right)^{n-1}$

Stirling Engine

→ 2 → η_{Carnot}

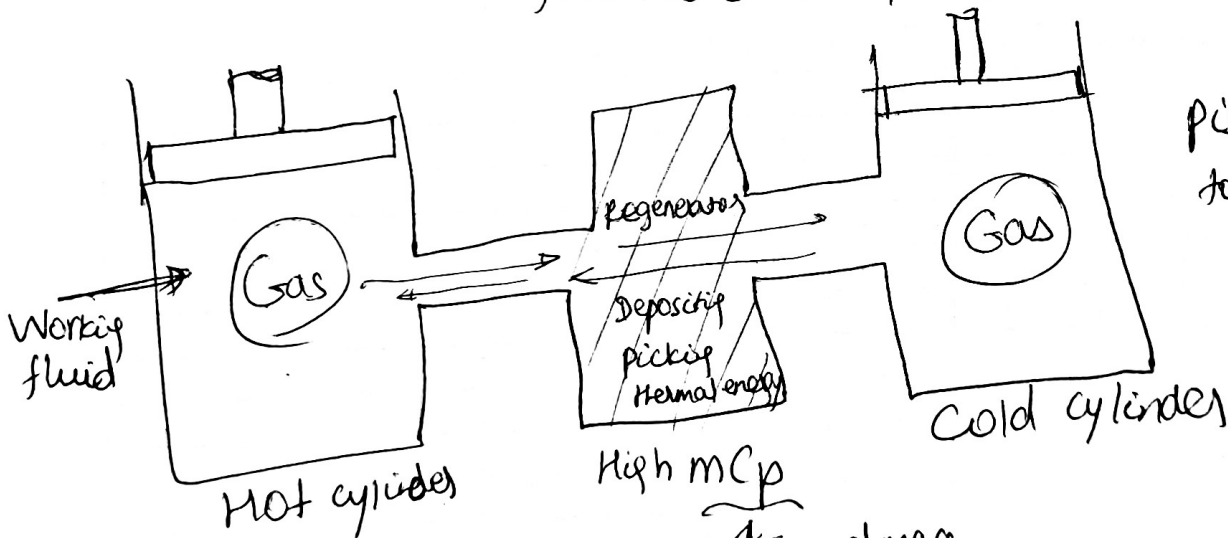


Pressure is going high:
compression → $\uparrow P$, $T \uparrow$
Rejecting heat
↳ will maintain T

Regenerator

takes thermal energy from one to another.

↳ from 2-3 to 4 to 1



Working fluid - Helium
Hydrogen